India 28 States: Plant Structure & Pesticides Guide

Comprehensive Agriculture, Plant Structure Anatomy & State-wise Pesticides Guide

1. Three Main Plant Structures (Plant Structure Anatomy)

From small herbs and shrubs to massive trees, all plants are fundamentally divided into 3 main anatomical parts:

I. Bottom Part - Root System

- **Primary Function:** Anchors the plant firmly in the soil and absorbs water and essential mineral nutrients (N, P, K) to send to upper parts.
- **Types:**
  - *Tap Root System:* Penetrates deep into the soil in trees and dicot plants (e.g., Mango, Neem).
  - *Fibrous Root System:* Spreads along the upper layer of soil in grasses, paddy, and small leafy greens.

II. Middle Part - Stem / Trunk System

- **Primary Function:** Provides structural support and strength. Transports water absorbed by roots and food prepared by leaves throughout the plant.
- **Vascular Tissues:** *Xylem* transports water and minerals upwards; *Phloem* transports prepared sugars/food downwards.
- **Structural Difference:** Thick woody bark provides outer protection in trees; soft green stems in smaller plants and herbs participate in photosynthesis.

III. Top Part - Leaves, Canopy & Organs

- **Primary Function:** Produces food via photosynthesis, handles transpiration, and manages gas exchange/respiration.
- **Components:** Leaves, terminal buds, flowers (reproduction), fruits, and seeds.

2. All 28 States of India - Agriculture & Pesticides Analysis

India comprises **28 states**. Below is the comprehensive state-wise mapping of major crops and commonly applied pesticides:

S.No	State	Major Crops	Commonly Used Pesticides
1	**Kerala**	Coconut, Rubber, Tea, Cardamom, Black Pepper, Paddy	Copper Oxychloride, Chlorpyrifos, Carbendazim
2	**Tamil Nadu**	Paddy, Sugarcane, Groundnut, Banana, Cotton, Maize	Imidacloprid, Cartap Hydrochloride, Chlorantraniliprole, Mancozeb
3	**Andhra Pradesh**	Chilli, Cotton, Paddy, Tobacco, Maize	Acephate, Thiamethoxam, Profenofos, Diafenthiuron
4	**Telangana**	Cotton, Paddy, Maize, Chilli, Turmeric	Fipronil, Spinetoram, Monocrotophos
5	**Karnataka**		

S.No	State	Major Crops	Commonly Used Pesticides
		Coffee, Ragi, Paddy, Sugarcane, Sunflower, Pigeon Pea	Hexaconazole, Propiconazole, Cypermethrin
6	**Goa**	Cashew, Coconut, Paddy, Arecanut	Bordeaux Mixture, Carbaryl, Glyphosate
7	**Maharashtra**	Sugarcane, Cotton, Soybean, Grapes, Onion	Dimethoate, Sulphur, Emamectin Benzoate, Lambda-cyhalothrin
8	**Gujarat**	Cotton, Groundnut, Castor, Cumin	Acetamiprid, Diafenthiuron, Profenofos
9	**Rajasthan**	Mustard, Pearl Millet (Bajra), Green Gram, Cumin, Coriander	Malathion, Quinalphos, Dimethoate
10	**Punjab**	Wheat, Paddy (Basmati), Cotton, Potato	Pretilachlor, Tricyclazole, Glyphosate, Thiamethoxam
11	**Haryana**	Wheat, Paddy, Mustard, Sugarcane, Cotton	Chlorpyrifos, Carbendazim, Imidacloprid
12	**Himachal Pradesh**	Apple, Potato, Maize, Vegetables	Mancozeb, Diphos, Dodine, Lime Sulphur
13	**Uttarakhand**	Paddy, Wheat, Sugarcane, Fruits, Pulses	Copper Fungicides, Cypermethrin, Carbaryl
14	**Uttar Pradesh**	Sugarcane, Wheat, Potato, Mustard, Mango	Carbofuran, Chlorpyrifos, Mancozeb
15	**Bihar**	Paddy, Maize, Wheat, Jute, Litchi	Phorate, Imidacloprid, Mancozeb
16	**Jharkhand**	Paddy, Pulses, Maize, Vegetables	Chlorpyrifos, Carbendazim, Cypermethrin
17	**West Bengal**	Paddy, Jute, Tea, Potato, Mango	Cartap Hydrochloride, Hexaconazole, Neem Formulations
18	**Odisha**	Paddy, Pulses, Groundnut, Cashew, Sugarcane	Triazophos, Imidacloprid, Carbendazim
19	**Chhattisgarh**	Paddy, Maize, Oilseeds, Vegetables	Cartap, Chlorpyrifos, Tricyclazole
20	**Madhya Pradesh**	Soybean, Wheat, Chickpea, Mustard	Chlorantraniliprole, Thiamethoxam, Tebuconazole
21	**Sikkim**	Cardamom, Ginger, Orange, Buckwheat (100% Organic State)	Bio-pesticides (Neem oil, Trichoderma, Vermicompost)
22	**Assam**	Tea, Paddy, Jute, Mustard	Copper Oxychloride, Ethion, Propargite
23	**Arunachal Pradesh**	Paddy, Maize, Orange, Pineapple, Ginger	Neem Oil, Copper Hydroxide, Bio-fungicides
24	**Nagaland**	Paddy, Maize, King Chilli, Cardamom	Organic Bio-pesticides, Lime Sulphur
25	**Manipur**	Paddy, Maize, Mustard, Fruits	Neem Formulations, Carbendazim
26	**Mizoram**	Paddy, Ginger, Turmeric, Bamboo, Fruits	Bio-pesticides, Organic Compost Controls
27	**Tripura**	Paddy, Rubber, Tea, Pineapple, Bamboo	Copper Oxychloride, Chlorpyrifos
28	**Meghalaya**	Paddy, Potato, Lakadong Turmeric, Ginger	Organic Bio-pesticides, Bordeaux Mixture

3. Safety & Organic Alternatives

Banned / Restricted Pesticides in India:

Highly toxic chemicals harmful to human health and the environment—such as **Endosulfan, Phorate, Methyl Parathion, Dichlorvos, and Alachlor**—are banned or under strict regulation in India.

Organic / Bio-Pesticide Alternatives:

- **Neem Oil (Azadirachtin):** Natural control for sap-sucking insects and caterpillars.
- **Panchagavya & Herbal Extracts:** Boost crop immunity and act as natural pest repellents.
- **Bio-Control Agents:** *Trichoderma viride* (against root rot) and *Pseudomonas fluorescens* (against bacterial diseases).